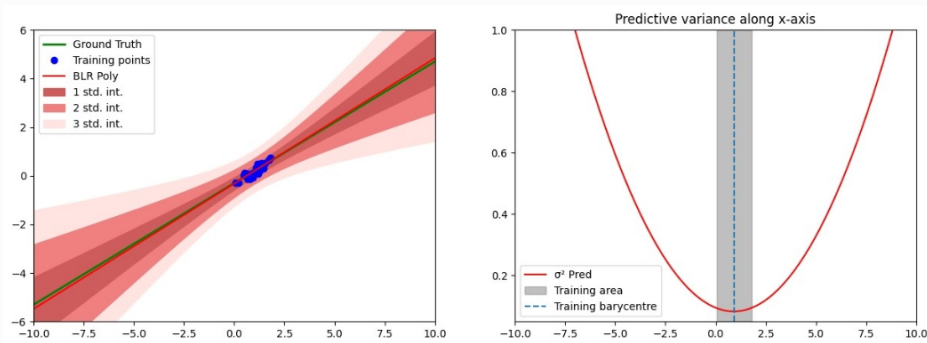


Robust Deep Learning Homework

Part 1: Bayesian Linear Regression

1.4



1.5

The predicted variance σ_{pred}^2 grows as the data x^* is far from the training area (the training points). In particular, it grows quadratically with the distance to the training area.

Intuitively, the further we are from the training data (i.e. from what we know) the less confident we are in our prediction, and thus the higher σ_{pred} is.

Let us prove it analytically, with $\alpha = 0$ and $\beta = 1$,

$$\Sigma_{\text{posterior}}^{-1} = \Phi^T \Phi = \begin{bmatrix} N & 1^T X \\ 1^T X & X^T X \end{bmatrix}$$

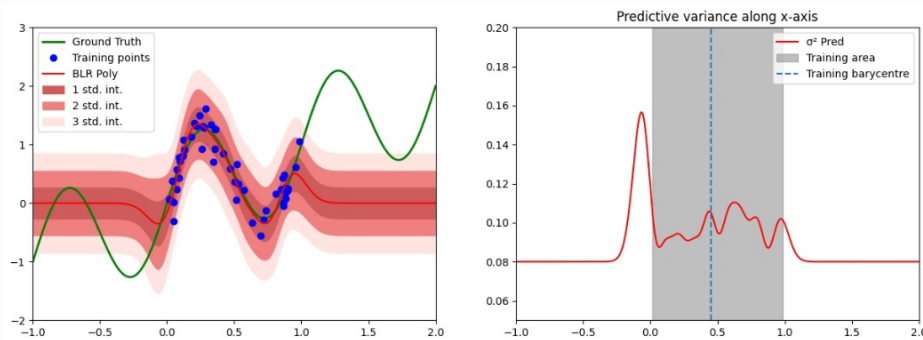
$$\Sigma_{\text{posterior}} = \frac{1}{N X^T X - (1^T X)^2} \begin{bmatrix} X^T X & -1^T X \\ -1^T X & N \end{bmatrix} = \frac{1}{|\Phi^T \Phi|} \begin{bmatrix} X^T X & -1^T X \\ -1^T X & N \end{bmatrix}$$

$$\begin{aligned} \sigma_{\text{prediction}}^2 &= 1 + \Phi(x^*)^T \Sigma_{\text{posterior}} \Phi(x^*) \\ &= 1 + \Phi(x^*)^T (\Phi^T \Phi)^{-1} \Phi(x^*) \\ &= 1 + \frac{1}{|\Phi^T \Phi|} \begin{bmatrix} 1 & x^* \end{bmatrix} \cdot \begin{bmatrix} X^T X & -1^T X \\ -1^T X & N \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x^* \end{bmatrix} \\ &= 1 + \frac{1}{|\Phi^T \Phi|} (X^T X - 2(1^T X) x^* + N x^{*2}) \end{aligned}$$

Thus, $\sigma_{\text{prediction}}^2$ is indeed a quadratic function of x^* , which is minimal in

$$x^* = \frac{2(1^T X)}{2N} = \frac{1}{N} \sum_{i=1}^N x_i \quad \text{the training barycentre.}$$

2.4 and 2.5



Now, the predictive variance stays quite small. It oscillates within the training area, is the largest near its border, and far from the training area, it quickly converges to $0.08 = \frac{1}{\beta}$.

When using localized basis functions like Gaussians, each basis function only has a significant influence over a limited region of the input space.

In regions far from the training data, these localized functions will all output values close to zero.

$$\text{Analytically, } \sigma_{\text{pred}}^2 = \frac{1}{\beta} + \Phi(x^*)^T \cdot \Sigma_{\text{pred}} \cdot \Phi(x^*) \rightarrow \frac{1}{\beta}$$

because $\Phi(x^*)$ tends quickly to 0 as x^* is outside the training area.

Part 2: Approximate Inference

1.2

Compared to the MAP estimate, the Laplace approximation provides a predictive prediction that is more nuanced. The level lines are curved away from the boundary between the two classes. The confidence of the points outside the training area is lower, which is desirable.

1.3

A larger weight decay imposes a stronger penalty on large weights. This encourages smaller weights, leading to a smoother and simpler decision boundary. A smaller weight decay would potentially lead to a more complex decision boundary, which could increase the risk of overfitting.

Moreover, a larger weight decay would imply a larger loss function Hessian, and thus a smaller Σ_{lap} . The Laplace model would be more certain about smaller weights.

1.4

The Linear Variational class implements a Bayesian linear layer using the reparametrization trick to sample weights from learnable Gaussian distributions. It also calculates the KL divergence between these weight posteriors and a Gaussian prior. The Variational Logistic Regression class simply uses this variational layer for its weights and applies a sigmoid output.

The main difference between Laplace and VI is that Laplace is a local Gaussian approximation of the posterior centered at the MAP estimate, while VI is a global optimization method that learns the parameters of an approximating distribution to minimize its KL with the true distribution.

1.5

The training loop for Variational Inference minimizes the negative ELBO. The ELBO loss is computed as the sum of the KL divergence and the binary cross-entropy loss. Compared to the MAP estimate, the predictive distribution from VI will show

uncertainty, represented by a less confident decision boundary outside of the training area. However, in the training area, VI is more confident than Laplace.